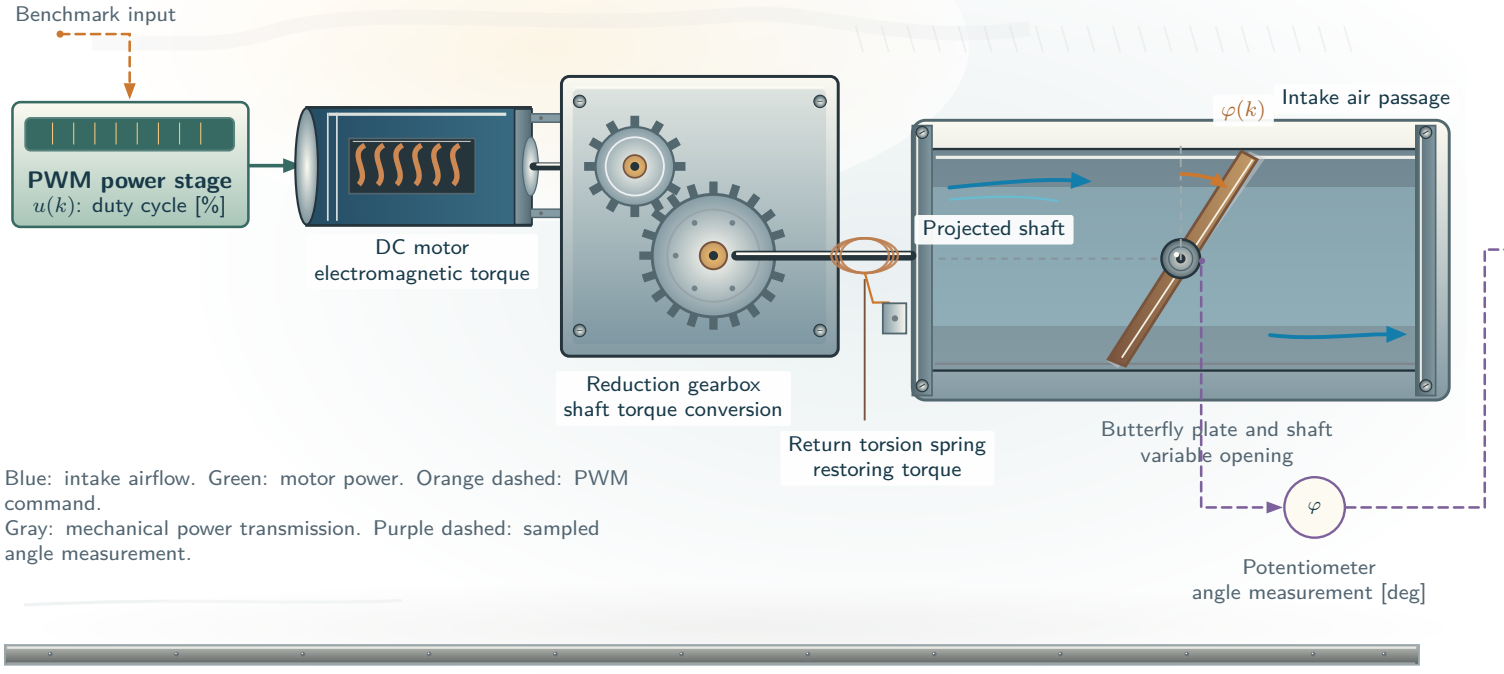
Automotive throttle: actuation and identification

Paper section 3.4: *Electro-mechanical throttle*

Physical chain: PWM drive, DC motor, gear box, spring and throttle plate



Blue: intake airflow. Green: motor power. Orange dashed: PWM command.
Gray: mechanical power transmission. Purple dashed: sampled angle measurement.

Automotive actuator / test-cell context (illustrative cutaway; not to scale)

Acquisition
Sample interval: $T_0 = 10$ ms.

Input: PWM duty [%].
Output: angle [deg].

Published plate range:
10° closed to 90° open.

Identification challenge
Friction, nonlinear spring and mechanical compliance can change the input–output response.

No calibrated internal parameters are inferred here.

Training experiment
Multisine excitation; $N = 10\,000$.
Fit dynamic models using the training record only. Keep measured duty and angle in their stated units.

Independent validation
Quasi-APRBS excitation; $N = 2\,500$.
Evaluate one-step and free-run prediction separately; report errors, residuals and model complexity.

Repository evidence
PWA/TS-inspired educational routines are available. Original measured time series have not been recovered; fixtures are synthetic.

Technology diagram, not a calibrated physical throttle model or an engineering assembly drawing. Gear ratio, electrical constants, spring law and friction map are intentionally unspecified. See THROTTLE_RECOVERY.md for the data-access record.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.